

Status or Strategy? Disentangling Peer Effects in Electric School Bus Adoption

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Abstract

The transition to zero-emission transportation is a critical public health and climate priority, yet adoption remains slow due to high upfront costs and technical uncertainty. Electric School Buses (ESBs), which cost three times more than diesel alternatives (\$375,000 vs. \$100,000) and require complex charging infrastructure, represent a high-stakes technology adoption decision for public school districts. While the U.S. Environmental Protection Agency (EPA) has committed \$5 billion to accelerate this transition through the Clean School Bus Program (CSBP), it remains unclear whether these initial subsidies catalyze broader adoption through local spillover effects or simply fund districts that would have adopted anyway. Understanding whether districts "learn from their neighbors" is crucial for designing subsidies that maximize the social multiplier of public funds.

Identifying peer effects is empirically challenging due to the "reflection problem": neighbors often behave similarly not because they influence each other, but because they share similar characteristics (e.g., climate, political preferences, or state incentives). To overcome this, I exploit a natural experiment created by the 2022 EPA Clean School Bus Program, which allocated over \$900 million in rebates via a randomized lottery. This design provides a "gold standard" exogenous shock: for a set of applicant districts, winning funding was determined by chance rather than underlying demand. I combine this lottery data with administrative records for the universe of U.S. school districts ($N \approx 13,000$) to estimate a Two-Stage Least Squares (2SLS) spatial model. I use a neighbor's random lottery win as an instrument for their adoption, allowing me to causally identify the effect of peer adoption on a focal district's decision to adopt.

Naive correlations suggest strong peer effects: districts are significantly more likely to adopt if their neighbors do. However, my causal analysis reveals a more nuanced story. Once I control for the non-random prioritization of low-income districts and isolate the variation driven by the lottery, the effect of "Emulation" (copying immediate geographic neighbors to "keep up with the Joneses") becomes statistically insignificant. In contrast, the effect of "Social Learning" (learning from regional peers who share similar operational challenges, such as climate and terrain) remains large and significant. Specifically, a 10% increase in adoption probability among regional climate peers increases a district's own adoption probability by approximately 1.5 percentage points.

These findings suggest that public sector technology adoption is driven primarily by information and risk reduction rather than social competition. School districts are sophisticated decision-makers who look to peers for relevant operational data (such as vehicle performance in cold weather) rather than simple visibility. For policymakers, this implies that the current strategy of scattering subsidies broadly to create "visibility" may be less effective than targeted "cluster" funding that generates dense, relevant information networks. Furthermore, the reliance on information transmission suggests that federal programs should invest heavily in technical assistance for early adopters to prevent negative experiences from discouraging future waves of adoption.

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